Consuming OData Services in SAP Fiori



Live Session Guidelines



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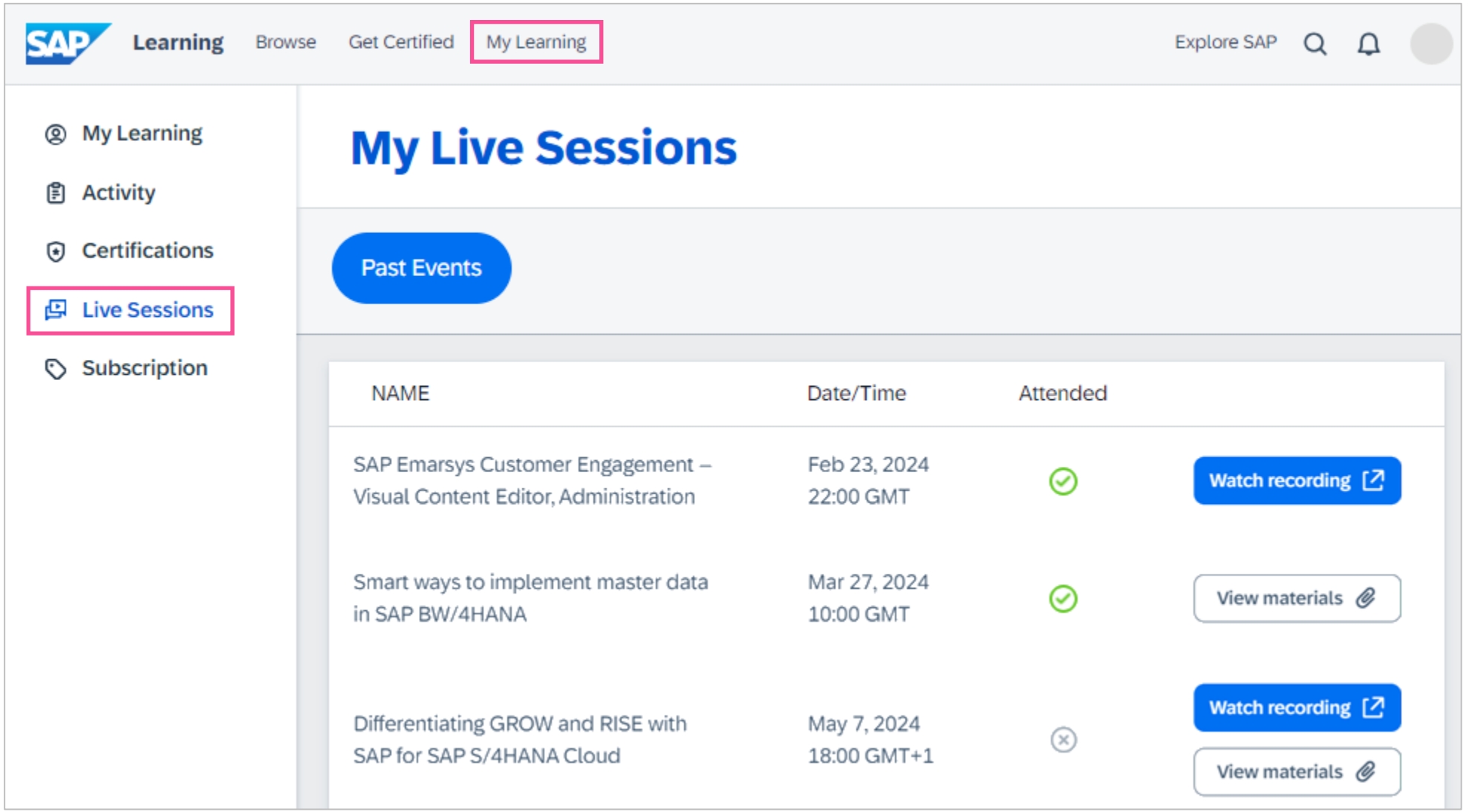


The live session is [not] being recorded.



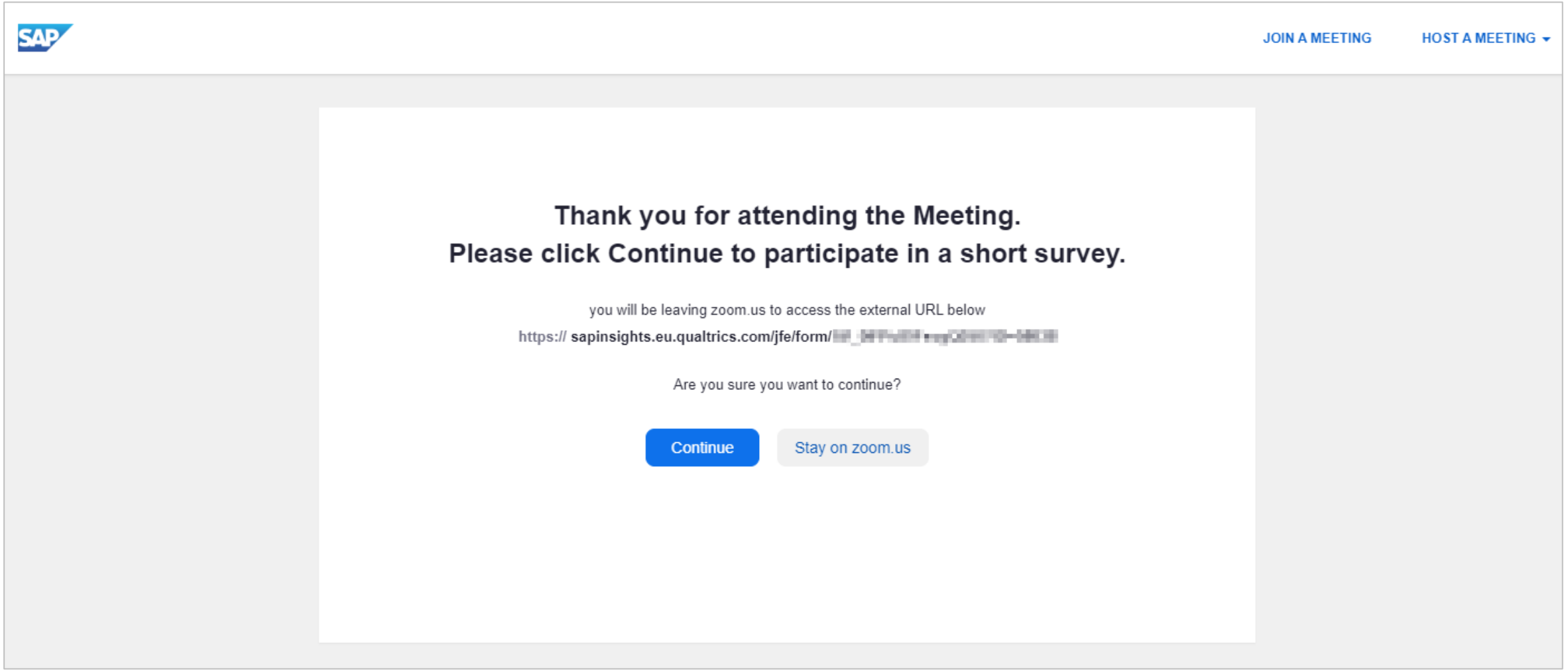
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Course Outline:



Understanding SAP FIORI

*Design Principles
*FIORI Architecture



SAP ODATA & FIORI

*History Context & Evolution of OData
*Key OData Concepts



ODATA Maintenance

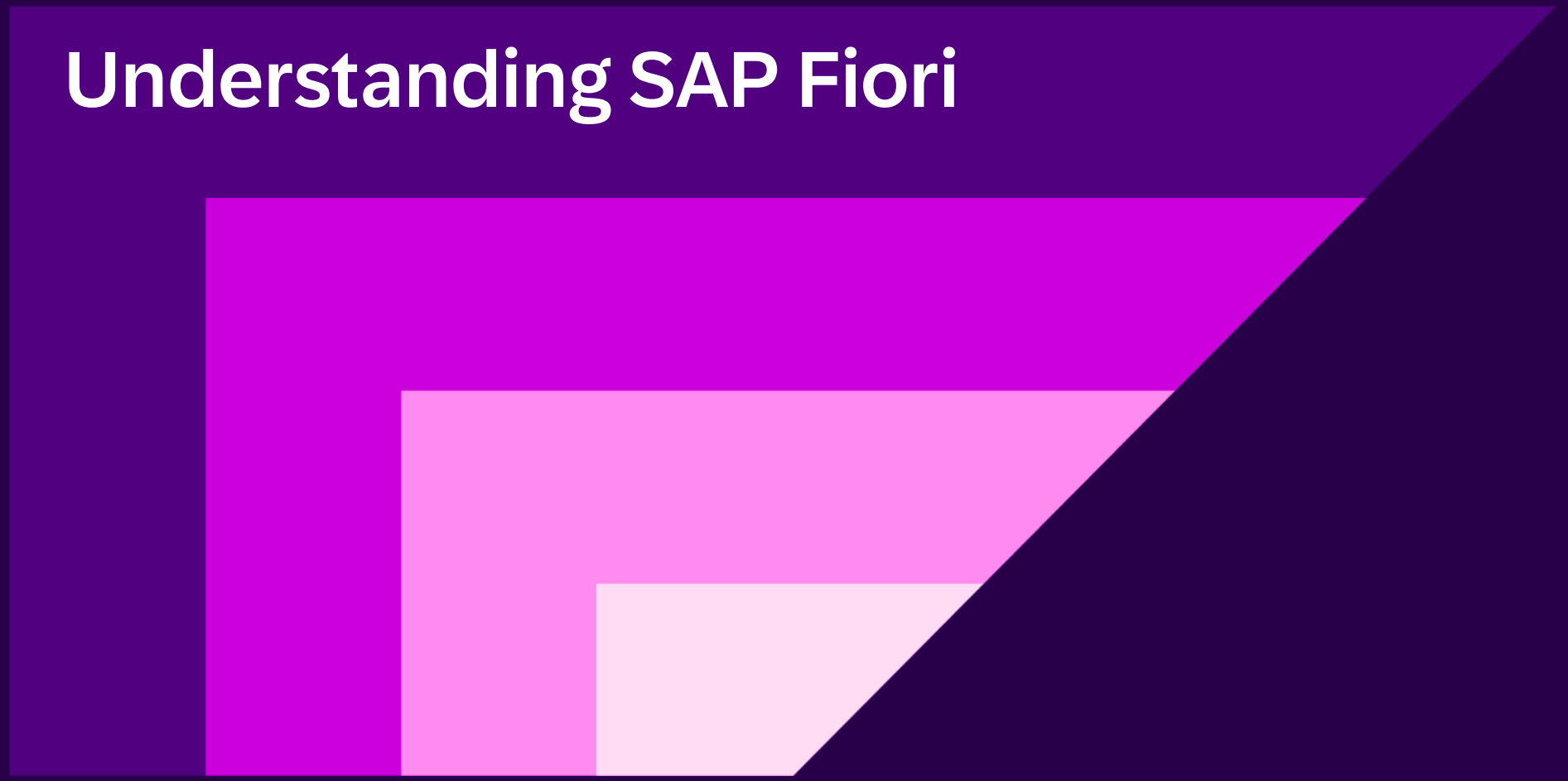
*XML vs JSON
*Debugging Option



RESTful Application Model & OData

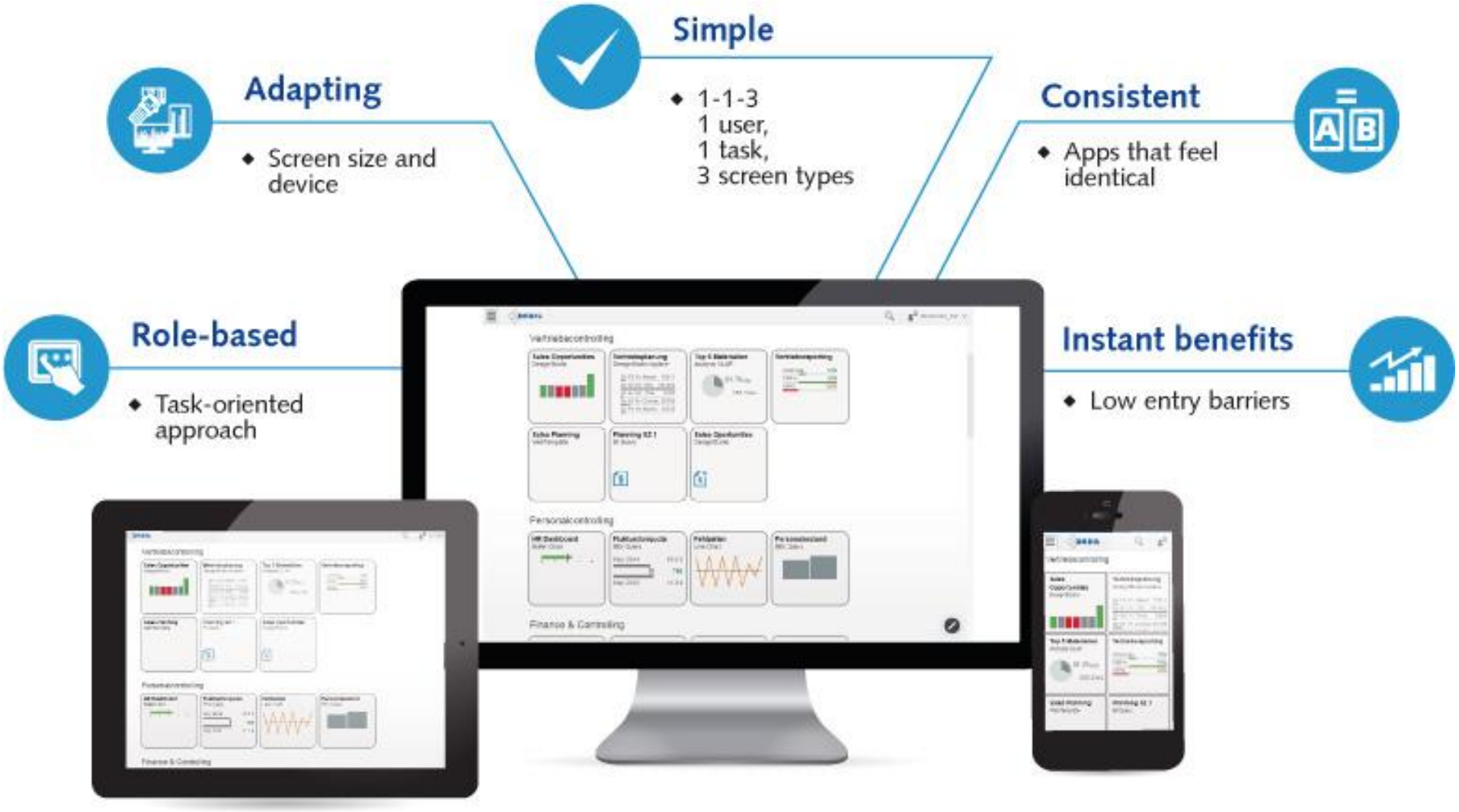
*CDS views
*CDS Views as OData Services

Understanding SAP Fiori



Public

Fiori Design Principles



Fiori Architecture & OData

SAP Fiori elements generates the app from OData services & annotations

What is displayed?

OData service

- Expose data & relations

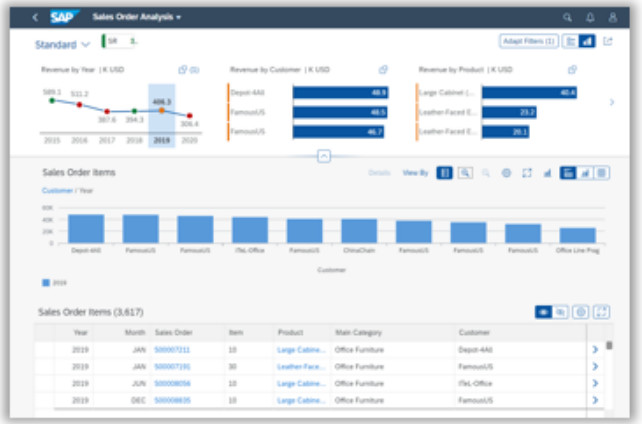
OData annotation

- Semantic metadata
- Behavior
- Consumption hints

Provided by:
Application Developers







How is it displayed?

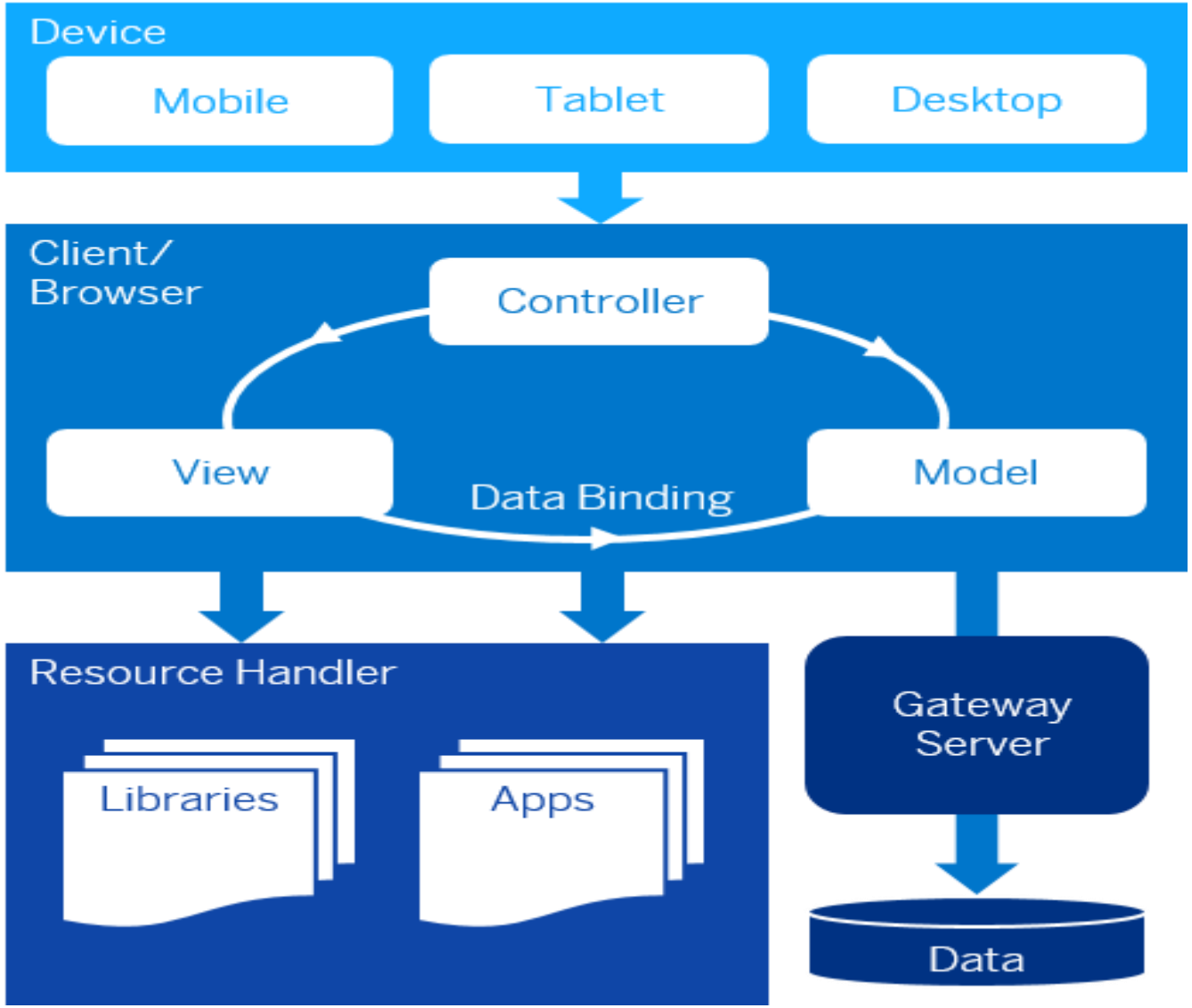
SAP Fiori elements floorplans

- Using SAPUI5 Controls
- Applying generic controller logic
- E.g. Analytical list page

Provided by:
SAP Fiori elements framework



MVC Model- Model View Controller

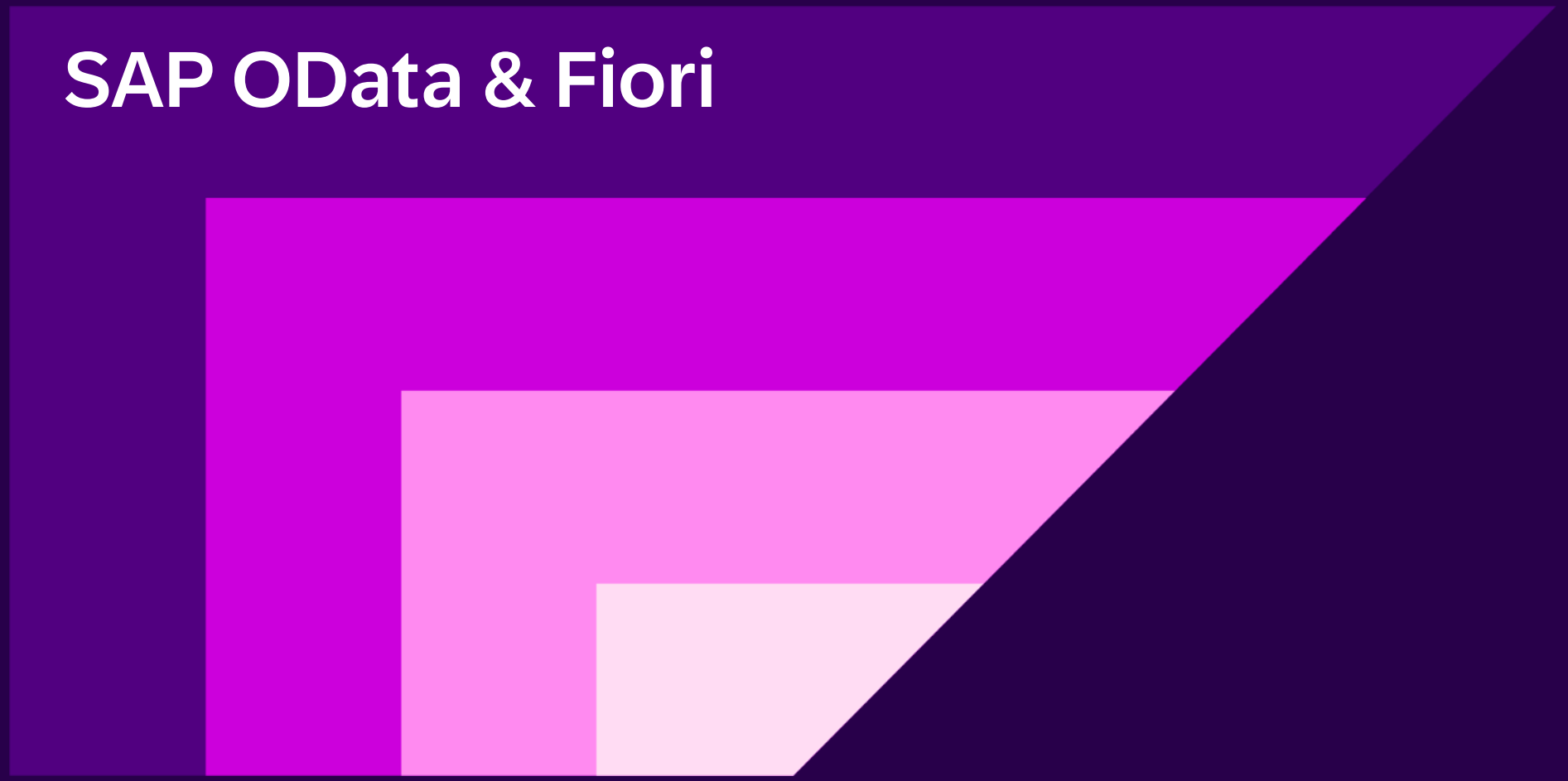




Demonstration 1:

Public

SAP OData & Fiori



Public

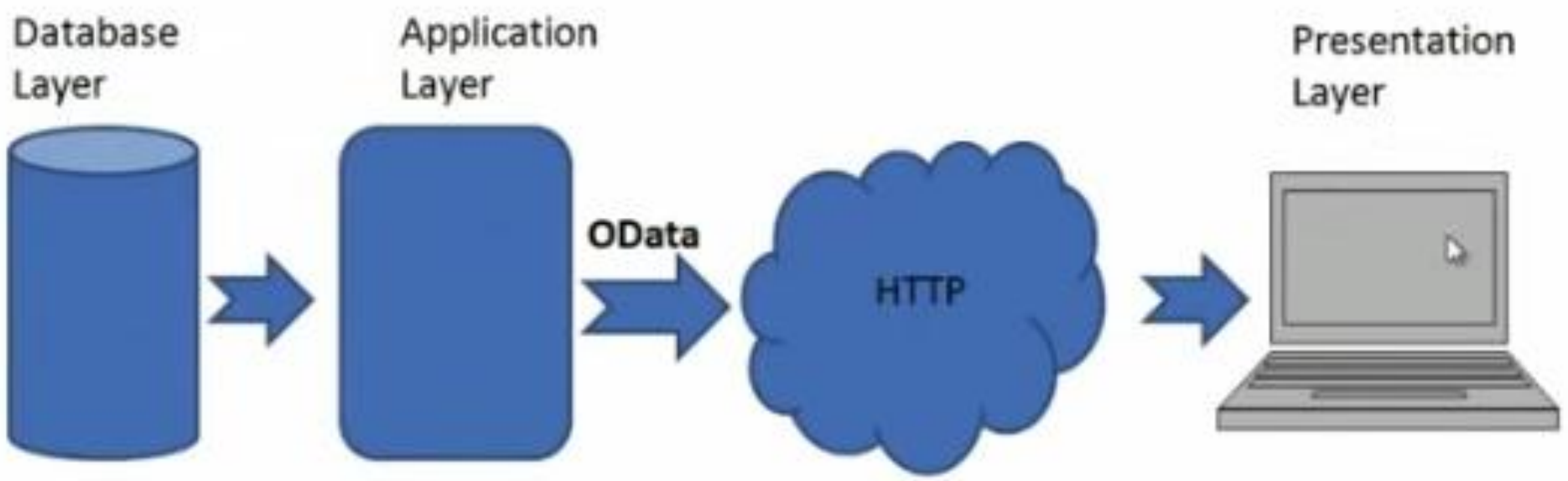
Historical Context and Evolution of OData

The Open Data Protocol (OData) addressed the need for a standardized protocol to query and update data across diverse systems. Utilizing web technologies like HTTP, AtomPub, and JSON, OData provided a consistent method for exposing and manipulating data using RESTful principles.

Over the years, OData evolved through several versions, enhancing functionality to meet changing developer and enterprise demands. Early versions laid the foundational concepts, while later versions introduced support for complex data types, request batching, and robust query options. Widespread adoption, including integration with SAP systems, marked Version 2.

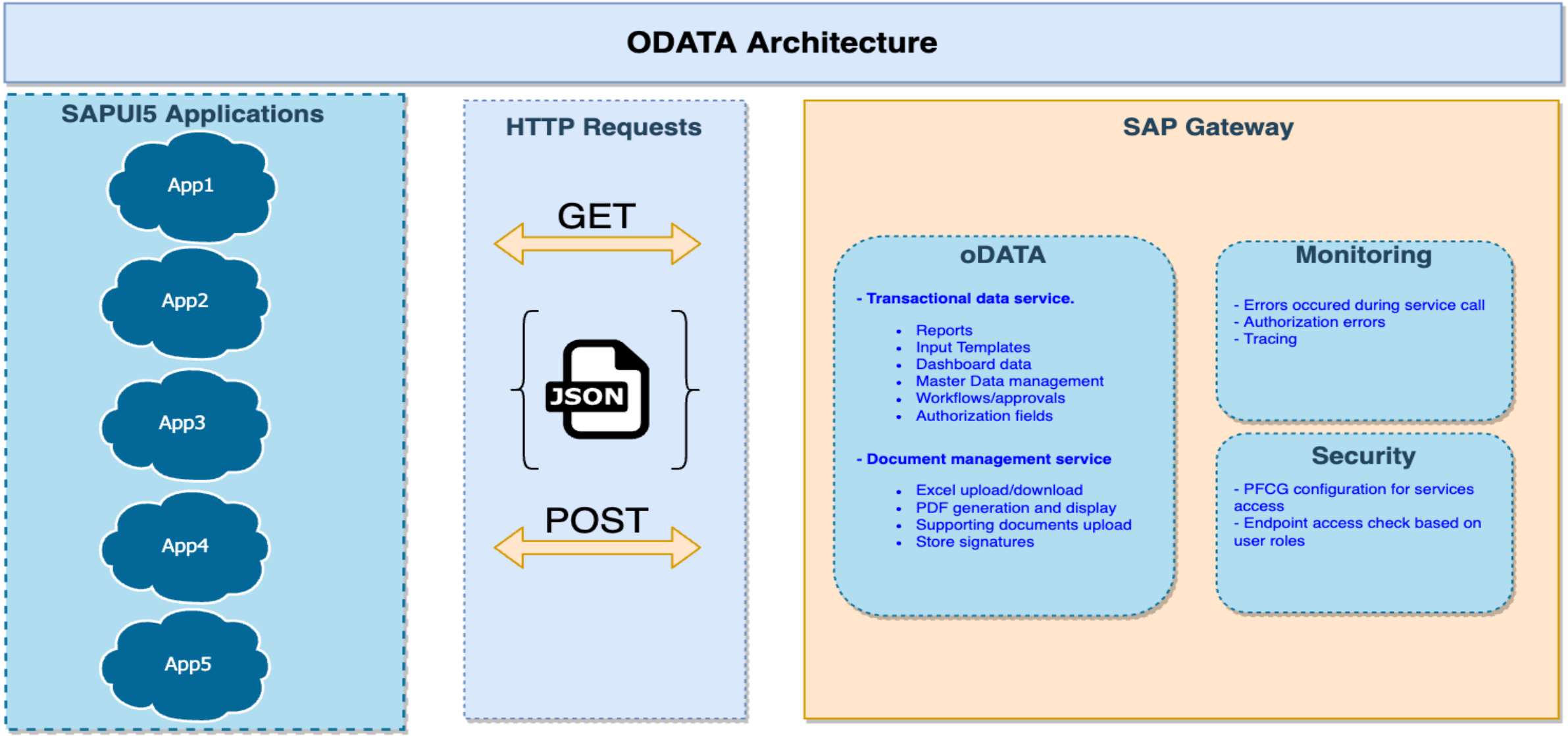
OData Version 4 brought significant improvements in performance, scalability, and flexibility, making it ideal for modern web and mobile applications. Understanding Odata's historical context and evolution highlights its adaptability and importance for seamless data exchange in contemporary enterprise systems.

Historical Context and Evolution of OData

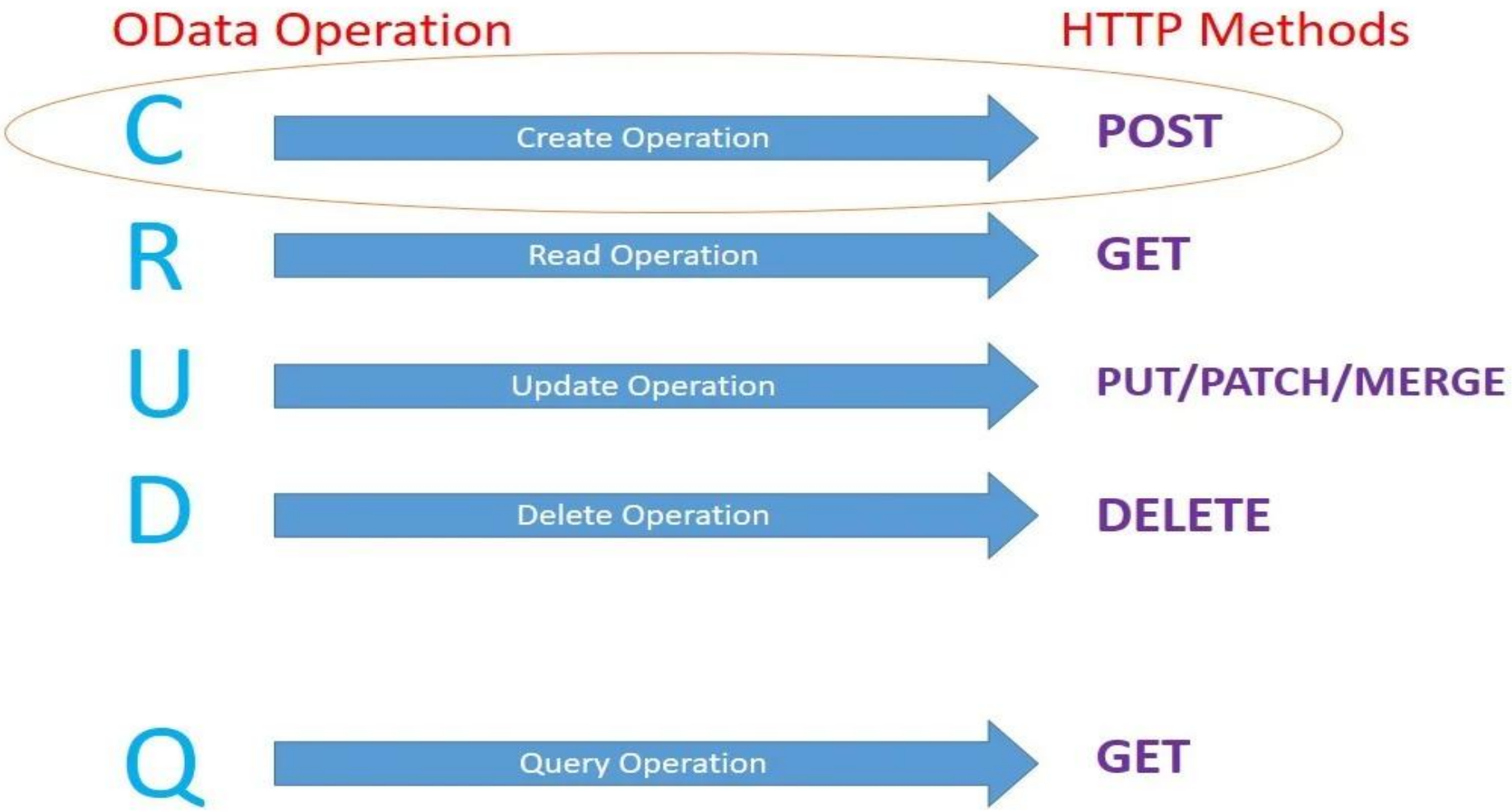


A Typical SAP Web Environment.

Key OData Concepts



OData Operations



Tools For Exploring OData Services



SAP Gateway Client — Test OData Services

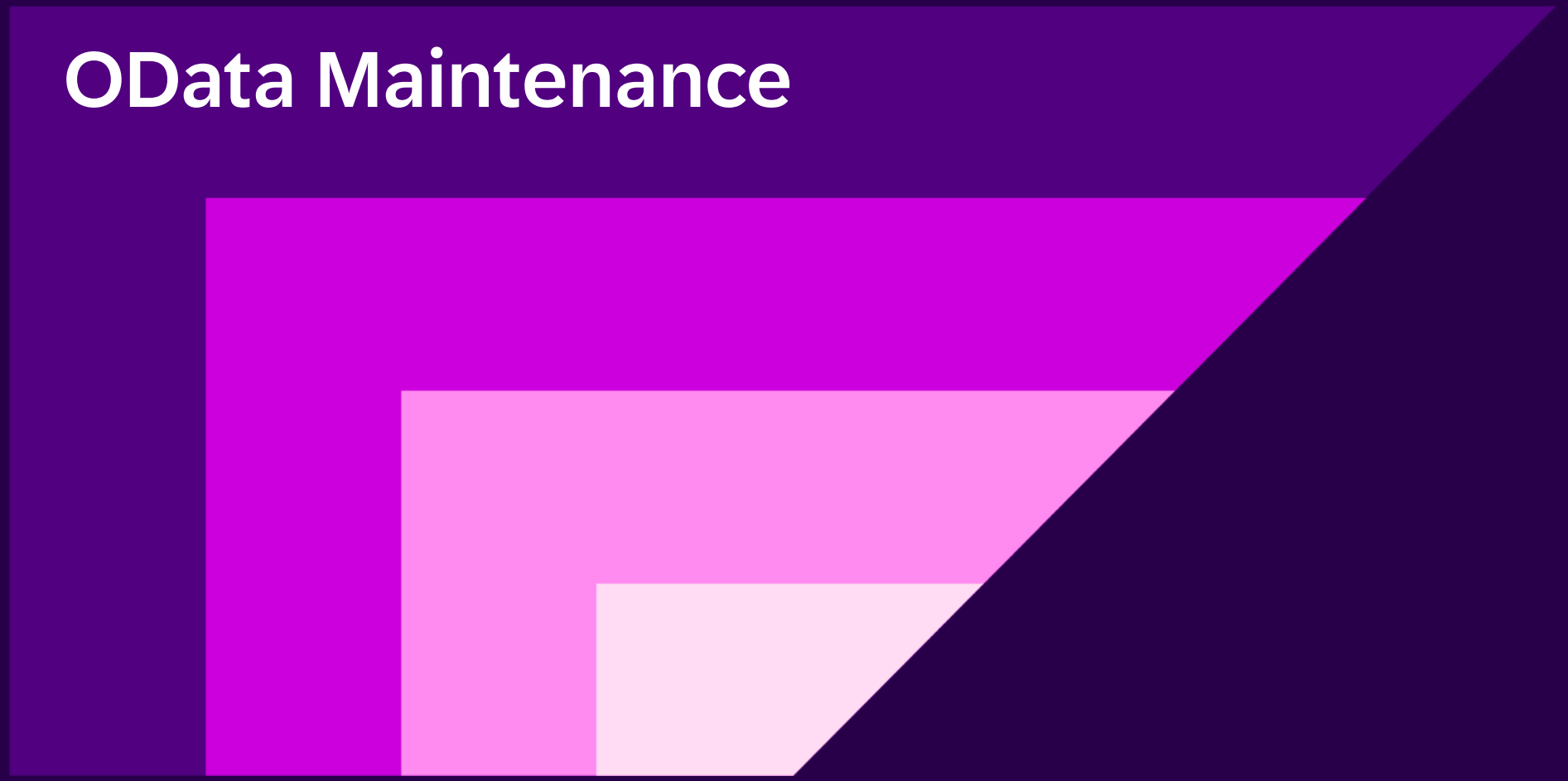


Browser Metadata — Service Structure via $metadata url



Postman — Simulate HTTP requests

OData Maintenance



Public

Atom versus JSON

XML (Atom)

```
▼<entry xmlns="http://www.w3.org/2005/Atom"
 xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata"
 xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices" m:etag="W/"datetime'2015-09-
 10T15%3A03%3A48.0000000'""
 xml:base="http://                              /sap/opu/odata/IWBEP/GWSAMPLE_BASIC/">
 ▶<id>...</id>
  <title type="text">ProductSet('HT-1000')</title>
  <updated>2015-09-10T15:22:03Z</updated>
  <category term="/IWBEP/GWSAMPLE_BASIC.Product"
  scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme"/>
  <link href="ProductSet('HT-1000')" rel="edit" title="Product"/>
  <link href="ProductSet('HT-1000')/ToSalesOrderLineItems"
  rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/ToSalesOrderLineItems"
  type="application/atom+xml;type=feed" title="ToSalesOrderLineItems"/>
  <link href="ProductSet('HT-1000')/ToSupplier"
  rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/ToSupplier"
  type="application/atom+xml;type=entry" title="ToSupplier"/>
 ▼<content type="application/xml">
  ▼<m:properties>
    <d:ProductID>HT-1000</d:ProductID>
    <d:TypeCode>PR</d:TypeCode>
    <d:Category>Notebooks</d:Category>
    <d:Name>Notebook Basic 15</d:Name>
    <d:NameLanguage>EN</d:NameLanguage>
   ▼<d:Description>
      Notebook Basic 15 with 2,80 GHz quad core, 15" LCD, 4 GB DDR3 RAM, 500 GB Hard Disc,
      Windows 8 Pro
    </d:Description>
    <d:DescriptionLanguage>EN</d:DescriptionLanguage>
    <d:SupplierID>0100000000</d:SupplierID>
    <d:SupplierName>SAP</d:SupplierName>
    <d:TaxTarifCode>1</d:TaxTarifCode>
    <d:MeasureUnit>EA</d:MeasureUnit>
    <d:WeightMeasure>4.200</d:WeightMeasure>
    <d:WeightUnit>KG</d:WeightUnit>
    <d:CurrencyCode>EUR</d:CurrencyCode>
    <d:Price>956.00</d:Price>
    <d:Width>0.30</d:Width>
    <d:Depth>0.18</d:Depth>
    <d:Height>0.03</d:Height>
    <d:DimUnit>M</d:DimUnit>
    <d:CreatedAt>2015-09-10T15:03:48.0000000</d:CreatedAt>
    <d:ChangedAt>2015-09-10T15:03:48.0000000</d:ChangedAt>
  </m:properties>
 </content>
</entry>
```

JSON

```
{
 - d: {
    + __metadata: {…},
      ProductID: "HT-1000",
      TypeCode: "PR",
      Category: "Notebooks",
      Name: "Notebook Basic 15",
      NameLanguage: "EN",
      Description: "Notebook Basic 15 with 2,80 GHz quad core,
      15" LCD, 4 GB DDR3 RAM, 500 GB Hard Disc, Windows 8 Pro",
      DescriptionLanguage: "EN",
      SupplierID: "0100000000",
      SupplierName: "SAP",
      TaxTarifCode: 1,
      MeasureUnit: "EA",
      WeightMeasure: "4.200",
      WeightUnit: "KG",
      CurrencyCode: "EUR",
      Price: "956.00",
      Width: "0.30",
      Depth: "0.18",
      Height: "0.03",
      DimUnit: "M",
      CreatedAt: "/Date(1441897428000)/",
      ChangedAt: "/Date(1441897428000)/",
    + ToSalesOrderLineItems: {…},
    + ToSupplier: {…}
   }
}
```

Debugging Query option

Body | **Request** | Response | Server | URI | Runtime

Request Line

```
GET /sap/opu/odata/iwbep/gwsample_basic/ProductSet?$format=json&sap-ds-debug=true HTTP/1.1
```

Request Header

Name	Value
host	s4dhost.wdf.sap.corp:44310
sec-ch-ua	"Not/A)Brand";v="99", "Google Chrome";v="115", "Chromium";v="115"
sec-ch-ua-mobile	?0
sec-ch-ua-platform	"Windows"
upgrade-insecure-requests	1
user-agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/:

Body | Request | **Response** | Server | URI | Runtime

Response Line

```
HTTP/1.1 200 OK
```

Response Header

Name	Value
SAP-Metadata-Last-Modified	Wed, 24 Mar 2021 17:02:30 GMT
Cache-Control	no-store, no-cache
Content-Type	application/json; charset=utf-8
DataServiceVersion	2.0

Body | Request | Response | **Server** | URI | Runtime

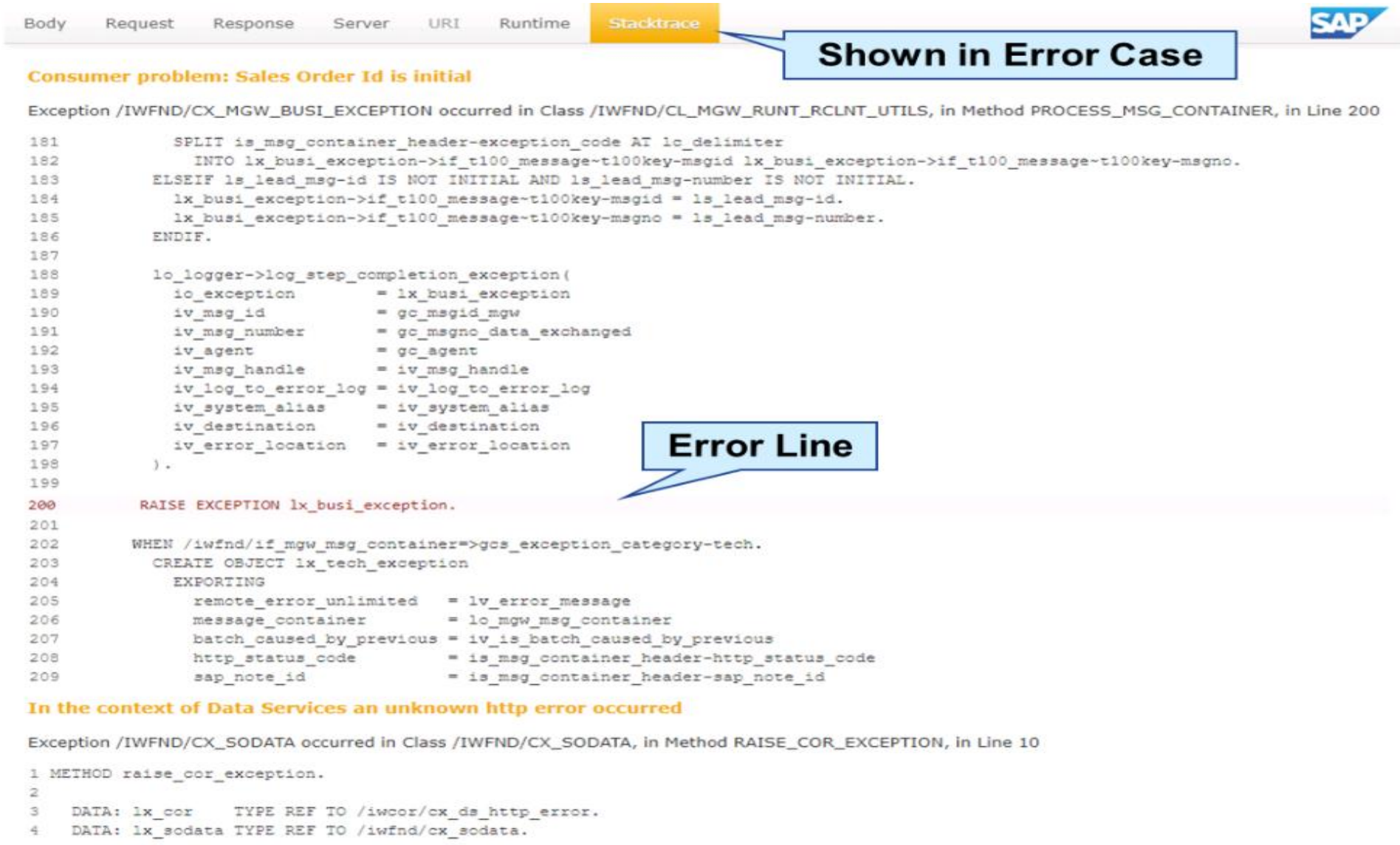
Server Environment

Name	Value
~request_line	GET /sap/opu/odata/iwbep/gwsample_basic/ProductSet?$format=json&sap-ds-debug=true HTTP/1.1
~request_method	GET
~request_uri	/sap/opu/odata/iwbep/gwsample_basic/ProductSet?$format=json&sap-ds-debug=true
~path	/sap/opu/odata/iwbep/gwsample_basic/ProductSet
~path_translated	/sap/opu/odata/iwbep/gwsample_basic/ProductSet
~query_string	$format=json&sap-ds-debug=true
~server_protocol	HTTP/1.1

Body | Request | Response | Server | URI | **Runtime**

```
└ /IWCOR/CL_DS_HDLR_HTTP->IF_HTTP_EXTENSION~HANDLE_REQUEST( ) 103.6
  └ /IWCOR/CL_DS_HDLR_ROOT->/IWCOR/IF_REST_HANDLER~HANDLE( ) 100.8
    └ /IWFND/CL_SODATA_ROOT_HANDLER->DISPATCH( ) 99.7
      ├ /IWFND/CL_MED_MDL_PROVIDER->GET_SERVICE_GROUP( ) 35.0
      ├ /IWFND/CL_SODATA_ROOT_HANDLER->HANDLE_AUTHORITY_CHECK( ) 0.0
      ├ /IWFND/CL_SODATA_SVC_FACTORY->CREATE_SERVICE( ) 1.2
      ├ /IWCOR/CL_DS_URI_FACADE->PARSE_URI( ) 2.7
      └ /IWFND/CL_SODATA_PROC_DISPTCHR->PROCESS( ) 40.5
        ├ /IWFND/CL_SODATA_PROCESSOR->/IWCOR/IF_DS_PROC_ENTITY_SET~READ( ) 38.1
        │ └ /IWFND/CL_SODATA_PROCESSOR->/IWCOR/IF_DS_PROC_ENTITY_SET~READ( ) 38.0
        │   └ /IWFND/CL_MGW_CO_DEPLOYM_PROXY->/IWFND/IF_MGW_CORE_RUNTIME~READ_ENTITYSET( ) 32.2
        │     └ /IWBEP/CL_MGW_LOCAL_HANDLER->GET_ENTITY_SET( ) 29.2
        │       ├ /IWBEP/CL_MGW_MED_PROVIDER->GET_LAST_MODIFIED( ) 0.2
        │       └ /IWBEP/CL_GWSAMPLE_BAS_DPC_EXT-
        │         >/IWBEP/IF_MGW_APPL_SRV_RUNTIME~GET_ENTITYSET( ) 13.1
        └ /IWFND/CL_SODATA_PROC_DISPTCHR->WRITE_TO( ) 2.3
          └ /IWFND/CL_SODATA_JSON_PROVIDER->WRITE_TO( ) 0.1
```

Error Case-Stacktrace



OData Metadata Structure

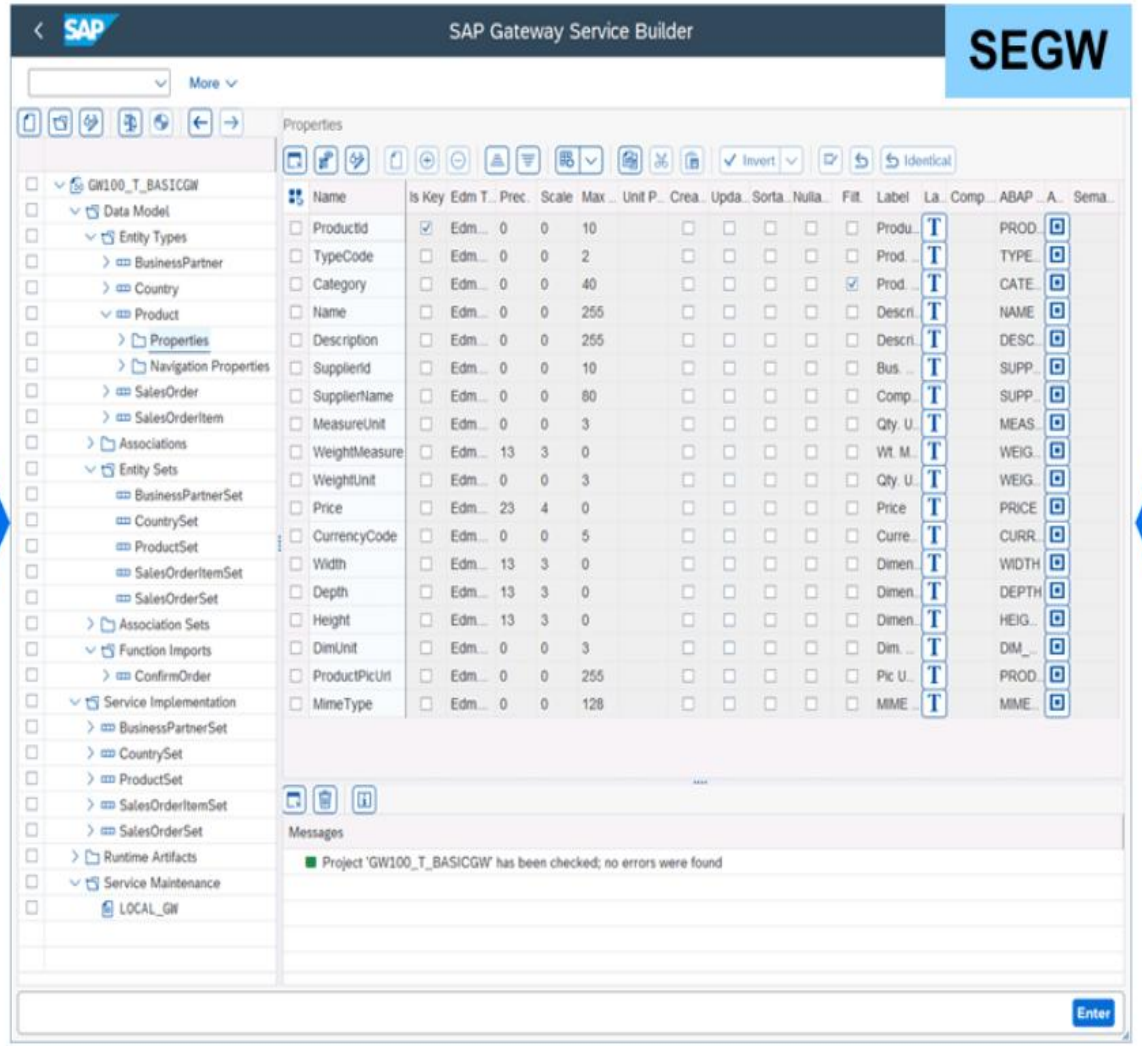
```
▼<edmx:Edmx xmlns:edmx="http://schemas.microsoft.com/ado/2007/06/edmx" xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata"
 xmlns:sap="http://www.sap.com/Protocols/SAPData" Version="1.0">
  ▼<edmx:DataServices m:DataServiceVersion="2.0">
    ▼<Schema xmlns="http://schemas.microsoft.com/ado/2008/09/edm" Namespace="GWSAMPLE_BASIC" xml:lang="en" sap:schema-version="1">
       <Annotation xmlns="http://docs.oasis-open.org/odata/ns/edm" Term="Core.SchemaVersion" String="1.0.0"/>
      ▶<EntityType Name="BusinessPartner" sap:content-version="1">
       ...
       </EntityType>
      ▼<EntityType Name="Product" sap:content-version="1">
        ▼<Key>
           <PropertyRef Name="ProductID"/>
         </Key>
         <Property Name="ProductID" Type="Edm.String" Nullable="false" MaxLength="10" sap:unicode="false" sap:label="Product ID" sap:updatable="false"/>
         <Property Name="TypeCode" Type="Edm.String" Nullable="false" MaxLength="2" sap:unicode="false" sap:label="Prod. Type Code"/>
         <Property Name="Category" Type="Edm.String" Nullable="false" MaxLength="40" sap:unicode="false" sap:label="Prod. Cat."/>
         <Property Name="Name" Type="Edm.String" Nullable="false" MaxLength="255" sap:unicode="false" sap:label="Product Name" sap:sortable="false" sap:filterable="false"/>
         <Property Name="NameLanguage" Type="Edm.String" MaxLength="2" sap:unicode="false" sap:label="Language" sap:creatable="false" sap:updatable="false"
         sap:sortable="false" sap:filterable="false"/>
         <Property Name="Description" Type="Edm.String" MaxLength="255" sap:unicode="false" sap:label="Prod.Descrip." sap:sortable="false" sap:filterable="false"/>
         <Property Name="DescriptionLanguage" Type="Edm.String" MaxLength="2" sap:unicode="false" sap:label="Language" sap:creatable="false" sap:updatable="false"
         sap:sortable="false" sap:filterable="false"/>
         <Property Name="SupplierID" Type="Edm.String" Nullable="false" MaxLength="10" sap:unicode="false" sap:label="Bus. Part. ID"/>
         <Property Name="SupplierName" Type="Edm.String" MaxLength="80" sap:unicode="false" sap:label="Company Name" sap:creatable="false" sap:updatable="false"/>
         <Property Name="TaxTarifCode" Type="Edm.Byte" Nullable="false" sap:unicode="false" sap:label="Prod. Tax Code"/>
         <Property Name="MeasureUnit" Type="Edm.String" Nullable="false" MaxLength="3" sap:unicode="false" sap:label="Qty. Unit" sap:semantics="unit-of-measure"/>
         <Property Name="WeightMeasure" Type="Edm.Decimal" Precision="13" Scale="3" sap:unicode="false" sap:unit="WeightUnit" sap:label="Wt. Measure"/>
         <Property Name="WeightUnit" Type="Edm.String" MaxLength="3" sap:unicode="false" sap:label="Qty. Unit" sap:semantics="unit-of-measure"/>
         <Property Name="CurrencyCode" Type="Edm.String" Nullable="false" MaxLength="5" sap:unicode="false" sap:label="Currency" sap:semantics="currency-code"/>
         <Property Name="Price" Type="Edm.Decimal" Precision="16" Scale="3" sap:unicode="false" sap:unit="CurrencyCode" sap:label="Unit Price"/>
         <Property Name="Width" Type="Edm.Decimal" Precision="13" Scale="3" sap:unicode="false" sap:unit="DimUnit" sap:label="Dimensions"/>
         <Property Name="Depth" Type="Edm.Decimal" Precision="13" Scale="3" sap:unicode="false" sap:unit="DimUnit" sap:label="Dimensions"/>
         <Property Name="Height" Type="Edm.Decimal" Precision="13" Scale="3" sap:unicode="false" sap:unit="DimUnit" sap:label="Dimensions"/>
         <Property Name="DimUnit" Type="Edm.String" MaxLength="3" sap:unicode="false" sap:label="Dim. Unit" sap:semantics="unit-of-measure"/>
         <Property Name="CreatedAt" Type="Edm.DateTime" Precision="7" sap:unicode="false" sap:label="Time Stamp" sap:creatable="false" sap:updatable="false"/>
         <Property Name="ChangedAt" Type="Edm.DateTime" Precision="7" ConcurrencyMode="Fixed" sap:unicode="false" sap:label="Time Stamp" sap:creatable="false"
         sap:updatable="false"/>
         <NavigationProperty Name="ToSupplier" Relationship="GWSAMPLE_BASIC.Assoc_BusinessPartner_Products" FromRole="ToRole_Assoc_BusinessPartner_Products"
         ToRole="FromRole_Assoc_BusinessPartner_Products"/>
         <NavigationProperty Name="ToSalesOrderLineItems" Relationship="GWSAMPLE_BASIC.Assoc_Product_SalesOrderLineItems"
         FromRole="FromRole_Assoc_Product_SalesOrderLineItems" ToRole="ToRole_Assoc_Product_SalesOrderLineItems"/>
       </EntityType>
```

/sap/opu/odata/iwbep/gwsample_basic/$metadata

SEGW-SAP GATEWAY SERVICE BUILDER



Application Skills

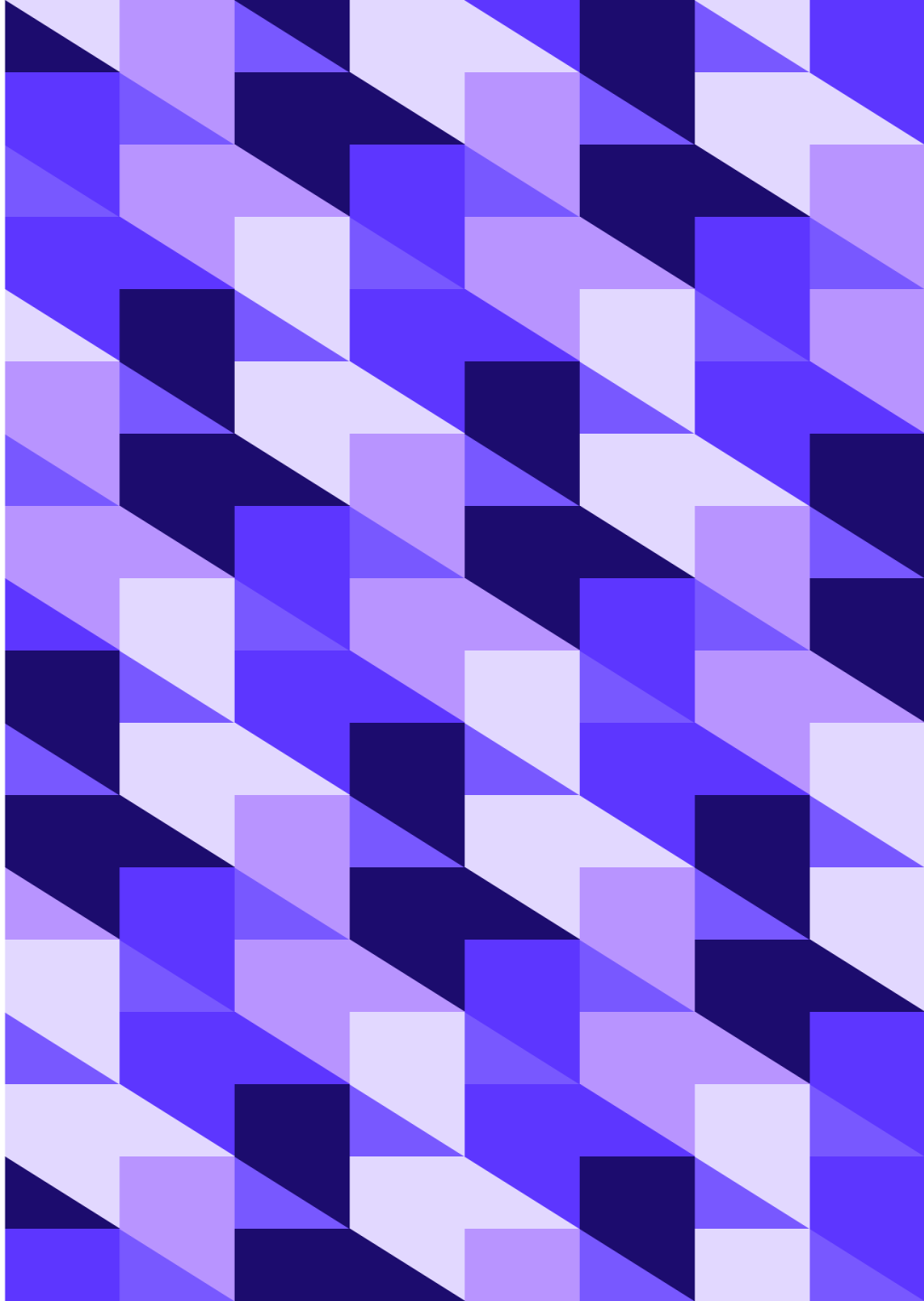




ABAP Skills



Demonstration 2:



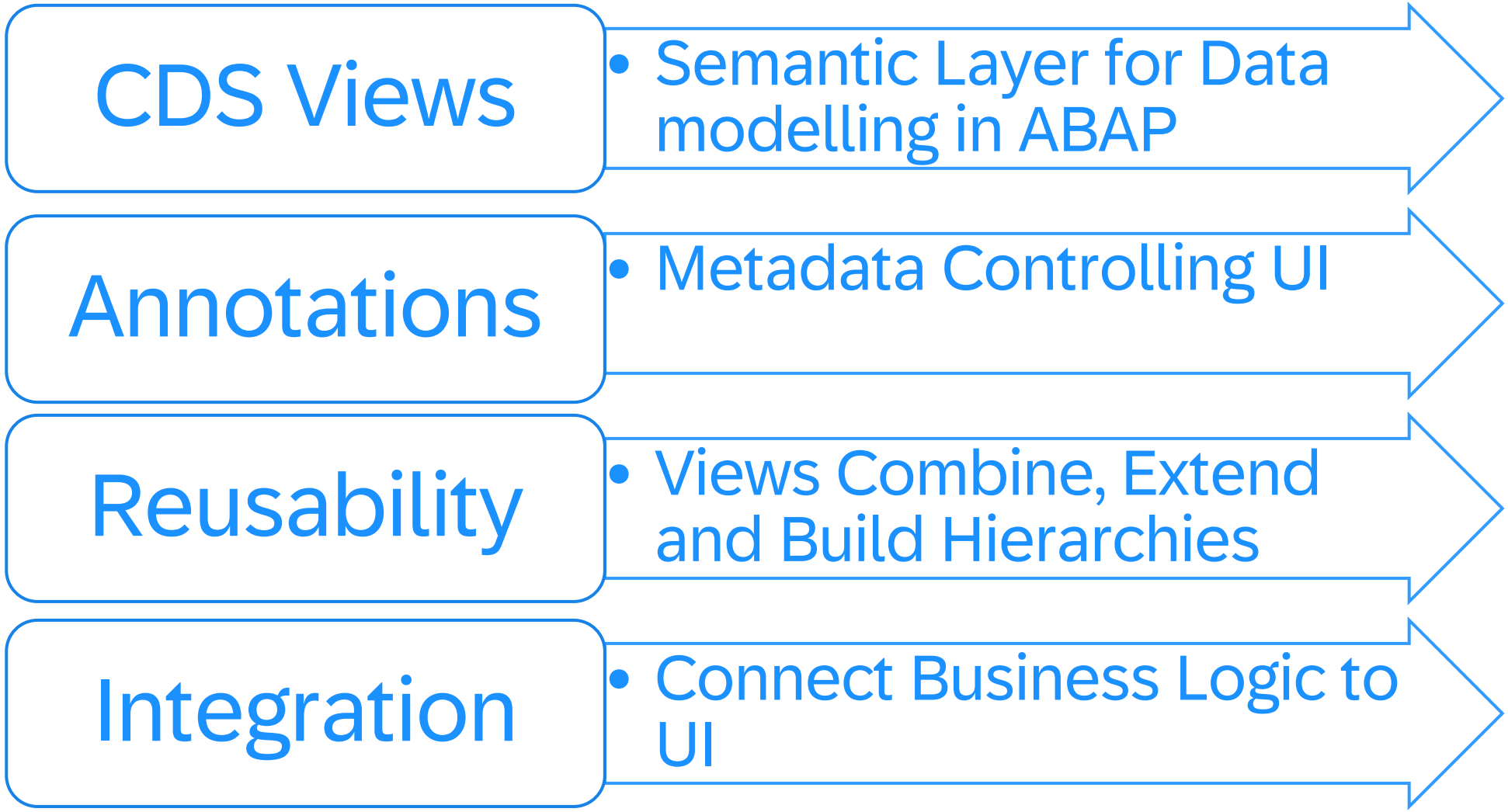
Public



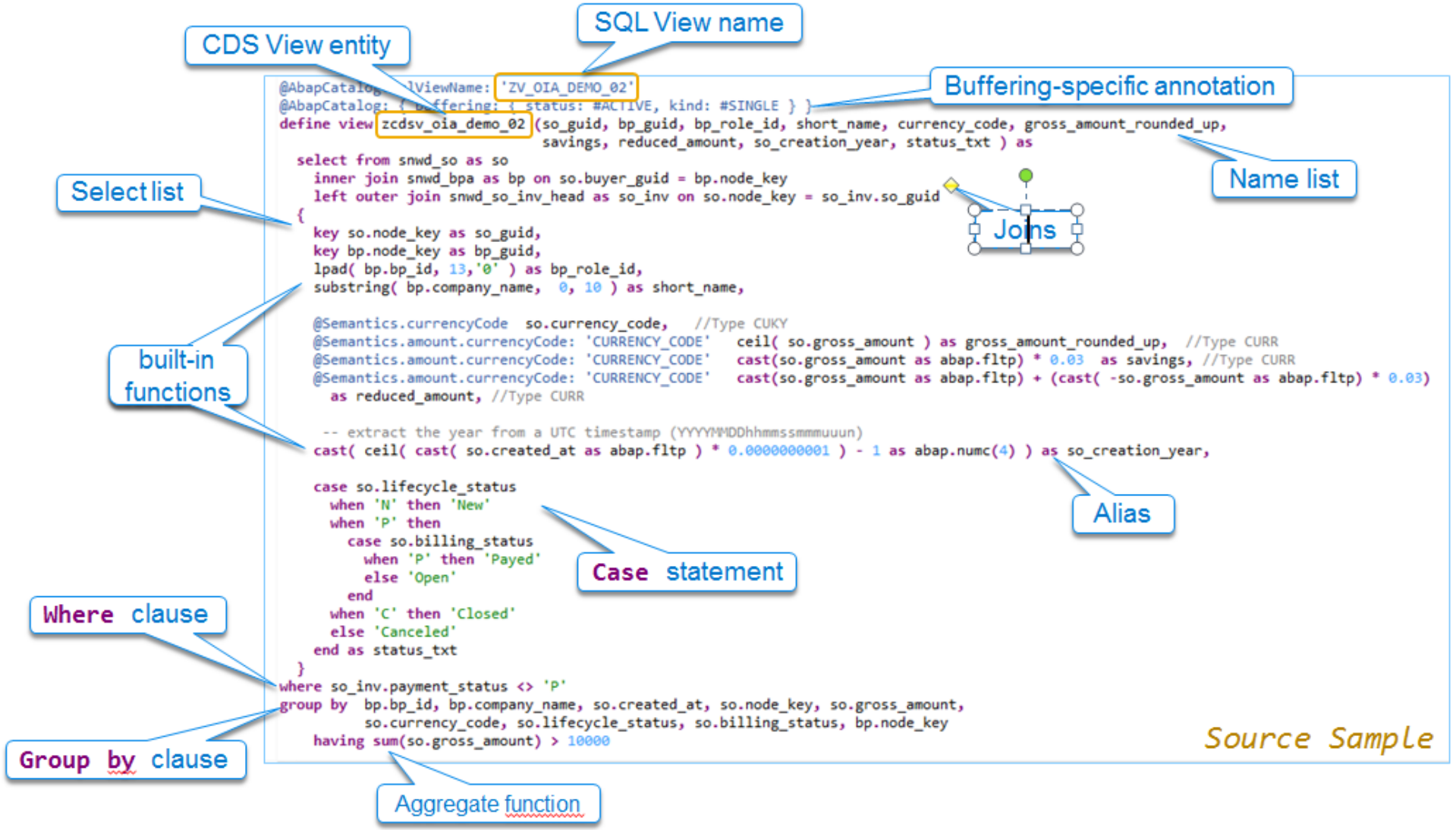
RESTful Application Model & OData Services

Public

What are CDS Views



Defining a CDS view



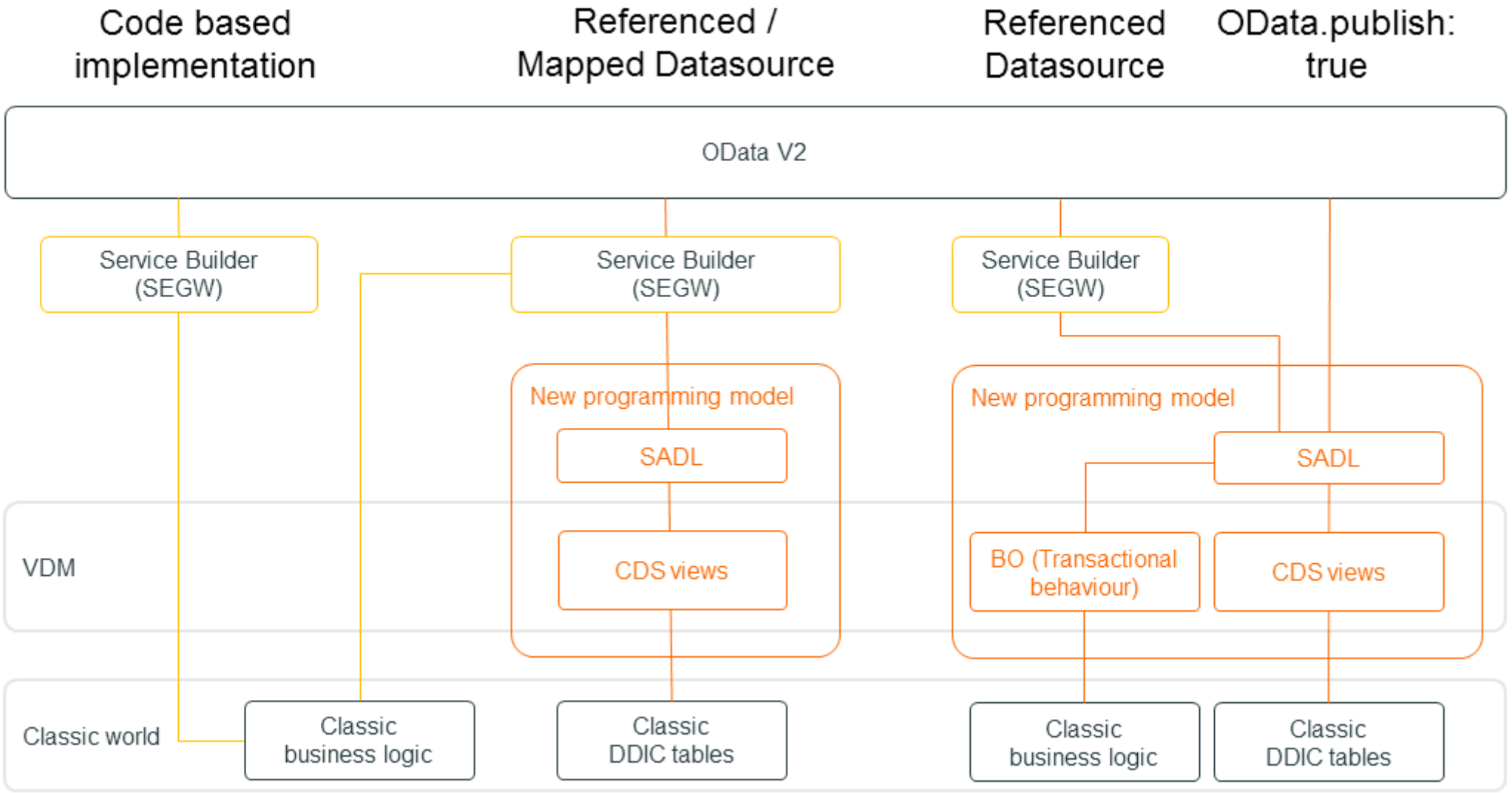

```
@AbapCatalog.sqlViewName: 'ZV_OIA_DEMO_02'
@AbapCatalog.buffering: { status: #ACTIVE, kind: #SINGLE }
define view zcdsv_oia_demo_02 (so_guid, bp_guid, bp_role_id, short_name, currency_code, gross_amount_rounded_up,
                               savings, reduced_amount, so_creation_year, status_txt ) as
  select from snwd_so as so
    inner join snwd_bpa as bp on so.buyer_guid = bp.node_key
    left outer join snwd_so_inv_head as so_inv on so.node_key = so_inv.so_guid
  {
    key so.node_key as so_guid,
    key bp.node_key as bp_guid,
    lpad( bp.bp_id, 13,'0' ) as bp_role_id,
    substring( bp.company_name,  0, 10 ) as short_name,

    @Semantics.currencyCode  so.currency_code,   //Type CUKY
    @Semantics.amount.currencyCode: 'CURRENCY_CODE'   ceil( so.gross_amount ) as gross_amount_rounded_up,  //Type CURR
    @Semantics.amount.currencyCode: 'CURRENCY_CODE'   cast(so.gross_amount as abap.fltp) * 0.03  as savings, //Type CURR
    @Semantics.amount.currencyCode: 'CURRENCY_CODE'   cast(so.gross_amount as abap.fltp) + (cast( -so.gross_amount as abap.fltp) * 0.03)
      as reduced_amount, //Type CURR

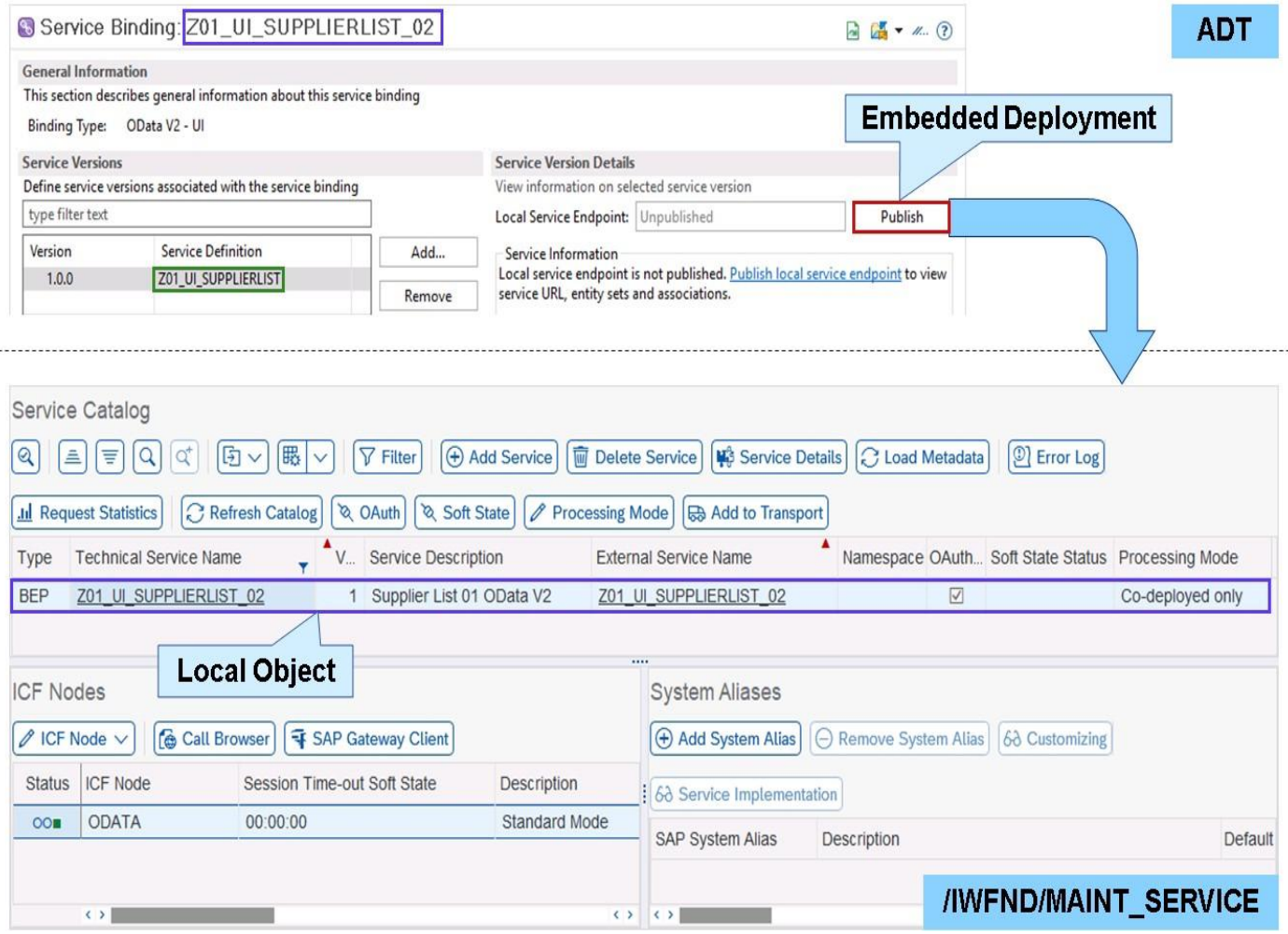
     -- extract the year from a UTC timestamp (YYYYMMDDhhmmssmmmuuun)
    cast( ceil( cast( so.created_at as abap.fltp ) * 0.0000000001 ) - 1 as abap.numc(4) ) as so_creation_year,

    case so.lifecycle_status
      when 'N' then 'New'
      when 'P' then
        case so.billing_status
          when 'P' then 'Payed'
          else 'Open'
        end
      when 'C' then 'Closed'
      else 'Canceled'
    end as status_txt
  }
  where so_inv.payment_status <> 'P'
  group by  bp.bp_id, bp.company_name, so.created_at, so.node_key, so.gross_amount,
            so.currency_code, so.lifecycle_status, so.billing_status, bp.node_key
    having sum(so.gross_amount) > 10000
```

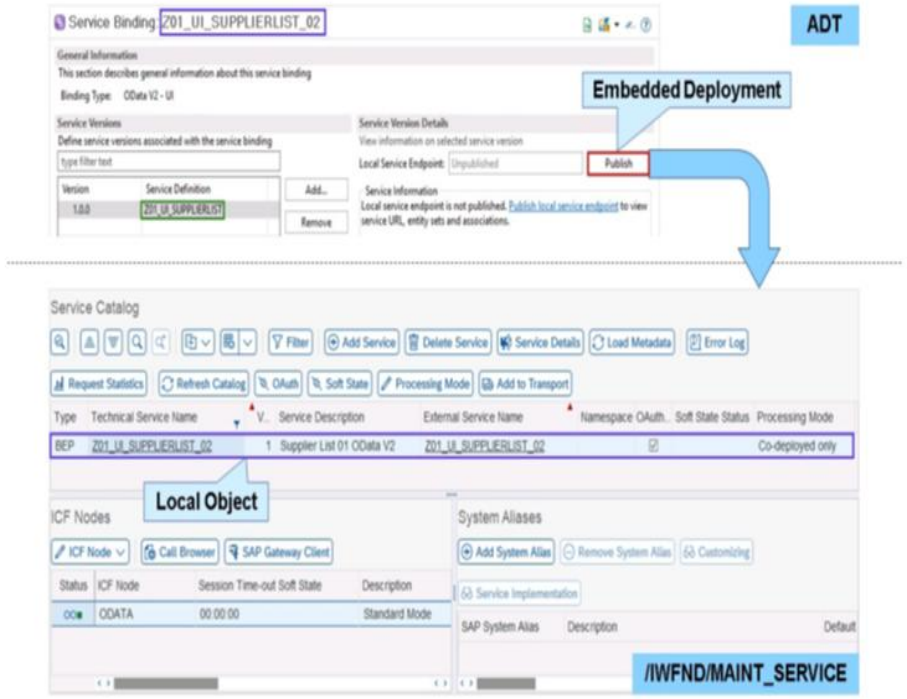
CDS Views as OData Sources

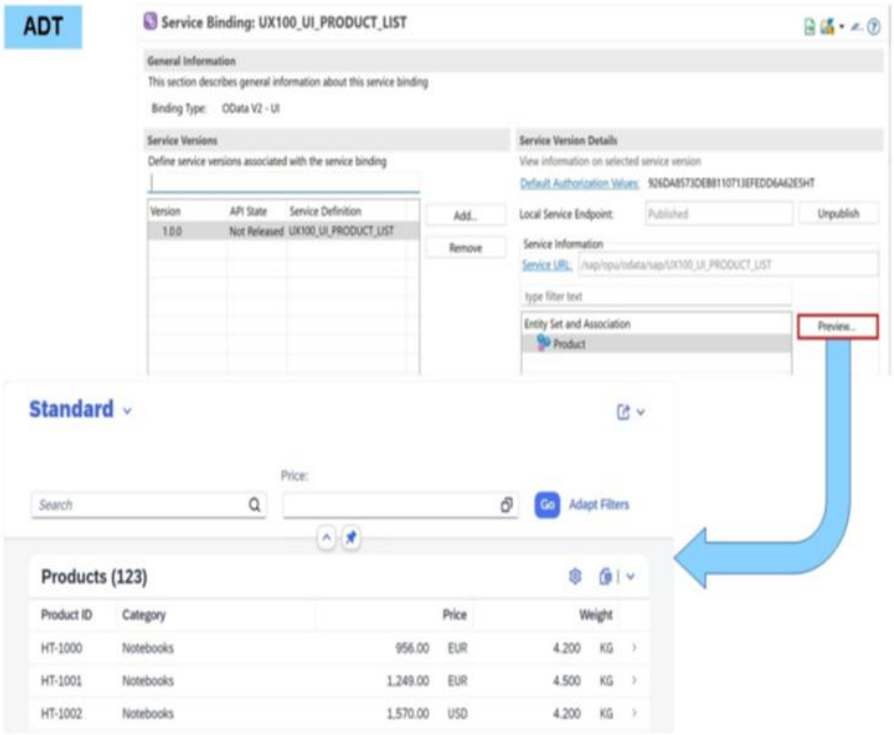


Consuming CDS Based OData in Fiori



Consuming CDS Based OData in Fiori





Demonstration 3

Public



Resources

Build an SAP Fiori App Using the ABAP RESTful Application Programming Model (RAP100)

Great mission covering CDS views → OData → Fiori app (includes elements and consumption):

https://developers.sap.com/mission.sap-fiori-abap-rap100.html

Create a Travel App with SAP Fiori Elements Based on OData V4 RAP Service

Hands-on group tutorial for Fiori elements + OData V4 consumption:

https://developers.sap.com/group.fiori-tools-odata-v4-travel.html

Get an Introduction to OData (good for history, evolution, key concepts):

https://developers.sap.com/group.scp-6-odata-intro.html

Create a List Report Object Page App with SAP Fiori Tools (Fiori elements basics + OData):

https://developers.sap.com/group.fiori-tools-lrop.html

Resources

Discussing the SAP Gateway and OData Services (excellent for Gateway, OData V2/V4 evolution, tools like SEGW):

https://learning.sap.com/learning-journeys/introducing-sap-abap-platform-fundamentals/discussing-the-sap-gateway-and-odata-services

Getting Started with SAP Fiori Elements: Understanding OData and Annotations

Directly covers OData + annotations + Fiori elements architecture:

https://learning.sap.com/courses/developing-an-sap-fiori-elements-app-based-on-a-cap-odata-v4-service/getting-started-with-sap-fiori-elements-understanding-odata-and-annotations_e936f375-fc74-4ed6-9820-11d3020f424c

Resources

Working With OData Annotations (detailed on semantic metadata, UI annotations for Fiori):

https://sapui5.hana.ondemand.com/sdk/docs/topics/83c89ccef12f48ab98f6c3811bd025b3.html

OData V4 Tutorial (modern consumption in SAPUI5/Fiori context):

https://sapui5.hana.ondemand.com/sdk#/topic/bcdbde6911bd4fc68fd435cf8e306ed0

How to Create an OData Service and Exposing CDS Views as an OData Service (practical steps):

https://blog.sap-press.com/how-to-create-an-odata-service-and-exposing-cds-views-as-an-odata-service

Resources – Live Sessions (Upcoming)

SAPUI5: Crafting a dynamic UI App with OData
https://learning.sap.com/live-sessions/sapui5-crafting-a-dynamic-ui-app-with-odata?searchId=88d1761a-7e97-4320-9bbc-b00e3d26096b&listPosition=2

SAP Fiori Spaces and Pages

https://learning.sap.com/live-sessions/sap-fiori-spaces-and-pages?searchId=0d67d09f-d50e-425d-a52d-590e42a3980e&listPosition=2

SAP Fiori Authorizations – Maintaining Catalogs

https://learning.sap.com/live-sessions/sap-fiori-authorizations-maintaining-catalogs?searchId=0d67d09f-d50e-425d-a52d-590e42a3980e&listPosition=4



Q & A

Thank you.



Public

Bring out your best.